

EXPERIEMNT 24

Prolog Program: Diet Suggestion System Based on Disease

Aim

To write a **Prolog program to suggest a suitable diet based on a person's disease** using facts and rules.

Algorithm

1. Define diseases as facts.
2. Define recommended food items for each disease.
3. Define foods to avoid for each disease.
4. Create rules to suggest the recommended diet.
5. Give the disease as a query.
6. Prolog displays the recommended foods and foods to avoid.

Prolog Code

% Diet Suggestion System Based on Disease

diet(diabetes, 'Vegetables, whole grains, pulses and low-sugar fruits').

diet(hypertension, 'Vegetables, fruits, whole grains and low-salt foods').

diet(anemia, 'Iron-rich foods, green leafy vegetables and beans').

diet(obesity, 'Vegetables, fruits, whole grains and low-fat foods').

diet(acidity, 'Bananas, oats, vegetables and light meals').

avoid(diabetes, 'Sugary drinks, sweets and excess refined carbohydrates').

avoid(hypertension, 'High-salt foods, processed foods and pickles').

avoid(anemia, 'Excess tea and coffee with meals').

avoid(obesity, 'Fried foods, sugary drinks and high-calorie snacks').

avoid(acidity, 'Spicy foods, excess coffee and carbonated drinks').

suggest(Disease) :-

diet(Disease, Diet),

avoid(Disease, Avoid),

write('Disease: '),

write(Disease),

nl,

write('Recommended Diet: '),

write(Diet),

nl,

write('Foods to Avoid: '),

write(Avoid),

nl.

Input

suggest(diabetes).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/expp-24.pl').  
compiling C:/Users/MOSHIYA/Downloads/expp-24.pl for byte code...  
C:/Users/MOSHIYA/Downloads/expp-24.pl compiled, 25 lines read - 2916 bytes written, 13 ms  
  
yes  
| ?- suggest(diabetes).  
Disease: diabetes  
Recommended Diet: Vegetables, whole grains, pulses and low-sugar fruits  
Foods to Avoid: Sugary drinks, sweets and excess refined carbohydrates  
  
yes  
| ?- |
```

Result

Thus, the diet suggestion system was successfully implemented using Prolog. The program suggests recommended foods and foods to avoid based on the given disease.